

PVTRACE · CALCULATION NOTE

PVTrace sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	sizing.py rerun for the capacitor bank selected at build (6.45 mF; R3 and R4 met on paper), a spare capacitor and the keypad and status LEDs (BOM line 18): cost USD 192, USD 27 over the target; mass 1,494 g, margin about 6 g; results.csv regenerated
0.4	2026-10-02	Amish Chadha	R3 row notes the capacitor selection rule and section 7 the accepted mass margin, decided on 2026-10-02 (PVT-DEC-001, items 3 and 4); no number changed
0.3	2026-10-01	Amish Chadha	Constructable design (PVT-DDR-003); mass and cost updated; R17 reported against the value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (sweep model, range, resolution, error budget, STC uncertainty, discharge, heat, battery, mass, cost) for the 6.6 mF isolated design in PVT-DDR-001

PVTrace sizing calculations

On paper, PVTrace meets ten of its seventeen requirements (R1 to R5, R10, R11, R12, R14, R15); R3 is met with the capacitor bank selected at build (see section 2). R17 (cost) is reported against the value-engineering target and is USD 27 over it. Two are **not met**: R13 (display readability in sun, not shown even with the sun hood, with heat at risk) and R16 (insulation screening, out of scope). R7 and R8 are **at risk** because they rest on the reference cell calibration, and R6 and R9 cannot be verified until hardware exists.

Three TRL 2 figures change. With the third capacitor (6.6 mF, PVT-DDR-001 decision 3) the high-current module sweeps in 20.5 ms at nominal capacitance but only 16.4 ms at the -20 % tolerance of electrolytic capacitors, so R3 was not met in the worst case as first drawn; Amish's decision of 2026-10-02 to select the bank at build changes that (section 2). The stored energy at 100 V rises from 22 J to 33 J. Battery life falls from about 16 h to 10.3 h once the isolation barrier (decision 4) and a 5 V boost are counted; R14 is still met. The module current at 99 % of Voc is up to 2.5 A, so the load switch must stay closed a few milliseconds past the end of the sweep to open below 0.5 A (R11).

Version 0.2 applies Amish's decisions of 2026-09-25 (PVT-DDR-002): the budget in `project.yaml` rises from \$150 to \$165, and a printed display sun hood (BOM line 15, +\$1, about 15 g) is added. The parts total goes from \$163 to \$164, so R17 changes from not met to met with a \$1 margin. Mass goes from 1.41 kg to 1.43 kg (R12 still met); the hood top is level with the isolator knob at 98 mm, so the envelope does not change. No other result changes.

Version 0.3 follows the constructable design of PVT-DDR-003: a 1.5 mm polycarbonate chassis plate, a USB-C charging socket, a smaller bonded window, a wider sun hood with screw tabs, a 16-bit ADC for the reference cell and added fixings. Mass goes from 1.43 kg to 1.49 kg (R12 still met, about 14 g to spare) and the estimated cost from USD 164 to USD 178. Following Amish's instruction of 2026-10-01, `budget_usd` (USD 165) is treated as a value-engineering target, not a limit, so R17 is reported as USD 13 over the target. The sweep, discharge, heat and battery results do not change.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the case dimensions from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates for a paper design; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Module cases	36-cell 100 W (22.0 V, 6.0 A); 108 half-cell 410 W (37.5 V, 13.9 A; MPP 31.4 V, 13.06 A); 144 half-cell 450 W reference (49.5 V, 11.6 A; MPP 41.5 V, 10.85 A); 2 x 60-cell 250 W in series (75.2 V, 8.9 A); rating limit, the reference curve scaled to 100 V and 20 A	Datasheet-class values, PVT-REQ-001
Module model	Single-diode model, ideality 1.2 per cell, series and shunt resistance solved from Isc, Voc, MPP and zero dP/dV at MPP	Standard five-parameter fit
Load capacitance	3 x 2200 μ F 160 V = 6.6 mF, ± 20 %	Decision 3, PVT-DDR-001; typical electrolytic tolerance
Loop resistance	56.6 m Ω : leads 8.6, MC4 contacts 2.0, fuse 5.0, isolator 2.0, shunt 4.0, MOSFET 10.0, capacitor ESR 20.0, wiring 5.0 m Ω	Copper at 20 °C; typical part values
Sampling	25,000 V-I pairs per second, voltage and current converted one after the other (20 μ s apart)	Half the rate of a 100 kS/s two-channel ADC
Front end	1 M Ω and 39 k Ω divider (0.1 %, 25 ppm/°C), 4 m Ω shunt (50 ppm/°C), gain 50, 4.096 V reference (50 ppm/°C), 12-bit ADC with ± 1 LSB INL, 30 °C between calibration and use	Typical parts in bom/ bom.csv line 4
Voc temperature coefficient	-0.30 %/°C; coldest cell -10 °C	Typical crystalline silicon
STC terms	Pmax coefficient -0.35 %/°C; module temperature ± 2 °C; spectral and angle mismatch ± 2 %; reference cell ± 3 to ± 5 %	PVT-PRC-001; decision 6
Discharge	22 Ω active dump, 10 k Ω passive bleed; one sweep every 5 s at most	PVT-PRC-001
Power	Controller and lit TFT 0.60 W; isolated side 0.25 W; 5 V boost 90 %; 3,000 mAh 18650 at 3.6 V, 90 % usable; 20 s of controller time per sweep	TRL 2 estimate plus isolation
Heat	9 W/(m ² K) from the case outside; 1,000 W/m ² on the lid; solar absorptance 0.30 (light grey) or 0.90 (dark); 45 °C ambient	Still air, natural convection plus radiation

■ 2. Sweep (R3, R4)

Table 2 gives the sweep for each case. The sweep time is the time for the capacitor to rise from 0 V to 99 % of Voc, integrated as $t = \int C dV / I(V)$ along the fitted curve, with the loop resistance and the bleed and divider currents included. It scales in proportion to the load capacitance.

Table 2. Sweep with 6.6 mF (single-diode model)

Case	Sweep, nominal (ms)	Sweep, -20 % C (ms)	Sweep, 4.4 mF (ms)	V-I pairs, nominal	Energy stored (J)	First sample (V)	Samples per volt near Isc, -20 % C
36-cell 100 W	28.9	23.1	19.3	722	1.6	0.34	22.0
108 half-cell 410 W (high current)	20.5	16.4	13.7	513	4.5	0.79	9.5
144 half-cell 450 W (reference)	32.2	25.8	21.5	805	7.9	0.66	11.4
2 x 60-cell 250 W (high voltage)	67.5	54.0	45.0	1,687	18.3	0.50	14.8
Rating limit, 100 V, 20 A	37.7	30.1	25.1	942	32.3	1.13	6.6

- **R3 with capacitors as bought is not met at worst-case tolerance.** The shortest sweep is 20.5 ms at nominal capacitance but 16.4 ms if the three capacitors are 20 % low. Holding 20 ms at -20 % would need 8.0 mF rated, which is a fourth capacitor. Amish decided on 2026-10-02 (PVT-DEC-001, item 4) not to add one but to measure each capacitor at build and fit the best three.
- **R3 with the selected bank.** The decided bank is 6.4 mF or more. On the high-current case 6.4 mF gives 19.9 ms, 0.1 ms short, and 20 ms needs 6.44 mF, so the calculation uses **6.45 mF** (proposed, awaiting Amish: the decided figure rounded up so the 20 ms bound holds). At 6.45 mF the high-current case sweeps in 20.05 ms, the shortest sweep of any case is 20.1 ms and the minimum is 501 V-I pairs, so **R3 and R4 are met on paper** with a bank selected to 6.45 mF or more. Three 2200 μ F parts need an average of 2150 μ F each, 2.3 % below rated, so BOM line 5 buys four and fits the best three (one spare). The longest sweep, the high-voltage case with +20 % capacitance, is about 81 ms, well inside 200 ms. The 4.4 mF column reproduces the TRL 2 estimates (13, 21 and 43 ms) within 2 ms, which checks the TRL 2 scaling.
- **R4 is met.** The fewest pairs are 410 (high-current case, -20 % capacitance), against 200.
- **Isc.** Loop resistance holds the module at 0.34 to 1.13 V at the first sample, 1 to 2 % of Voc, so firmware extrapolates Isc from the first samples. Pre-charging the bank negative is not needed.
- **Module capacitance.** With a capacitive load $dV/dt = I / C_{load}$, so the module's own capacitance C_{mod} causes a current error of about C_{mod} / C_{load} , independent of the sweep time. An error of 0.5 % or less needs C_{mod} of 26 μ F or less at the lowest load capacitance. The capacitance of TOPCon and HJT modules at MPP is not verified in this note; the 20 ms bound in R3 remains a working figure until a source or a measurement confirms it.
- **Sample skew.** Converting V and I 20 μ s apart gives up to 0.13 % error in Pmax at nominal capacitance (0.16 % at -20 %), in the high-current case. Firmware removes it by interpolating each voltage to the time of its current sample.

■ 3. Range, resolution and accuracy (R1, R2, R5, R6)

Range (R1, R2). Cold-weather Voc at a -10 °C cell temperature is 24.3 V, 41.4 V, 54.7 V and 83.1 V for the four cases, all below 100 V. Two 450 W reference modules in series reach 99.0 V at STC and 109.4 V at -10 °C, so the 100 V check refuses them; the manual must say so. At 100 V the capacitors have a 1.60 times voltage margin and the MOSFETs 1.50 times. At 20 A the shunt dissipates 1.6 W against its 3 W rating, and the load MOSFET conducts 0.15 J per sweep. Both are met by design review.

Resolution. The divider gives a voltage full scale of 109.1 V (26.6 mV per count); the shunt and amplifier give 20.48 A full scale (5.00 mA per count).

Accuracy (R5). After a two-point calibration, drift over 30 °C leaves ± 0.17 % of reading ± 0.024 % of full scale on voltage and ± 0.21 % of reading ± 0.026 % of full scale on current, inside ± 1 % ± 0.1 %. The amplifier offset before calibration is worth 6.25 mA, so the zero must be calibrated. R5 is met on paper; it depends on calibration against a meter and on the component grades in `bom/bom.csv` line 4.

Repeatability (R6). Quantization at the reference MPP is 0.079 % of Pmax per sample and the instrument contributes ± 0.27 % to Pmax, so the ± 1 % target will be set by irradiance drift between sweeps. R6 cannot be verified at TRL 3.

■ 4. Irradiance and STC uncertainty (R7, R8)

Table 3. STC Pmax uncertainty (root sum square)

Term	Value
Reference cell calibration	± 3 to ± 5 %
Spectral and angle mismatch	± 2 %
Module temperature, ± 2 °C at -0.35 %/°C	± 0.7 %
Instrument, V and I	± 0.27 %
Total	± 3.7 to ± 5.4 %

R8 (± 5 %) is met only if the reference cell calibration is within ± 4.5 %; R7 (± 5 % irradiance) needs ± 4.6 %. Both are **at risk** until the once-calibrated reference cell (decision 6) shows its uncertainty. CalRig (reference ± 0.2 °C from 10 to 40 °C) can check the temperature probe in that range but not at module operating temperatures of 60 °C and more, and not the reference cell.

■ 5. Discharge and heat (R11, R13)

Discharge (R11, met). The active path has a 145 ms time constant (174 ms at +20 % capacitance): 100 V falls below 30 V in 0.21 s and below 1 V in 0.80 s, against 2 s. The passive bleed has a 66 s time constant (79 s at +20 %): 100 V falls below 60 V in 40 s, against 60 s. Peak dump power is 455 W and peak dump current 4.5 A for milliseconds; the bleed draws 10 mA (1.0 W) at 100 V, which firmware subtracts from the current reading. At 99 % of Voc the module still delivers up to 2.51 A (rating limit case), more than the 0.5 A at which R11 allows the circuit to open. Firmware must keep the load switch closed until the current falls below 0.5 A, at most 6 ms more with +20 % capacitance. With that rule R11 is met.

Stored energy. 33 J at 100 V (22 J with 4.4 mF). At one sweep every 5 s the dump resistor averages 1.6 W for the reference module and 6.6 W at the 100 V limit.

Heat (R13, at risk). The case has 0.113 m² of outside area and a 0.0286 m² lid, so it sheds 1.02 W/K. In full sun at 45 °C ambient:

Table 4. Inside temperature, lumped model

Case color	Reference module, one sweep per 5 s	100 V limit, one sweep per 5 s
Light grey (absorptance 0.30)	11.0 W, about 56 °C	16.0 W, about 61 °C
Dark (absorptance 0.90)	28.2 W, about 73 °C	33.2 W, about 78 °C

Solar gain on the lid dominates. Even a light case reaches about 56 °C inside, above the usual 45 °C charging limit of a Li-ion cell and close to the limits of commodity display boards. The design therefore needs a light-colored case, a charger with a temperature cut-off (added to `bom/bom.csv` line 10), and the advice to keep the tracer in shade between sweeps. A printed sun hood over the display is now fitted (PVT-DDR-002 item 12). It shades the window on three sides and from above; its small shading of the lid is ignored here, so the figures above stay conservative. Display readability in direct sun is still **not met** on paper: a hood improves contrast but readability of the adopted TFT (decision 5) can only be shown in a field check, which is TRL 4 work.

■ 6. Battery (R14)

The controller and display draw 0.60 W and the isolated side 0.25 W at 5 V, or 0.94 W from the cell through the boost. A 3,000 mAh cell with 90 % usable (9.7 Wh) gives **10.3 h**, about 1,853 sweeps at 20 s each. R14 (8 h or 200 sweeps) is met. The TRL 2 figure of 16 h left out the isolation and the boost.

■ 7. Mass, size and cost (R12, R17)

Table 5. Mass estimate

Item	Mass (g)
Enclosure body and lid, ABS 3 mm, with bumpers and glands	402
Load capacitors, 3 x 75 g	225
Test leads, 2 x 1 m 4 mm ² with MC4	162
Sensor pod with 3 m cable	150
DC isolator	120
Battery, holder, charger	72
Hardware and consumables	50
Controller and display	45
Dump and bleed resistors	45
Measurement board; MOSFETs and bar; fuse and holder	40 each
Isolation board	10
Display sun hood, PETG, with screws	17
Window, PC 72 x 54 x 3 mm, and gasket	16
Chassis plate, PC 1.5 mm (model volume 37.8 cm ³)	45
USB-C charging socket with lead	8
Total	1,486

The window was 10 g in v0.2, too low for 3 mm polycarbonate; it is now computed from the model (16 g).

R12 is met: 1.49 kg (1,494 g with the keypad and LEDs, 8 g) against 1.5 kg, with about 6 g to spare on estimated masses (was 14 g) (accepted by Amish on 2026-10-02, with the prototype weighed at TRL 4; PVT-DEC-001, item 3). Over the bumpers the case is 234 x 144 x 80 mm, 244 mm over the glands and 98 mm to the top of the isolator knob and sun hood, inside 250 x 150 x 100 mm.

The priced BOM totals **USD 192.00**. Value-engineering target: USD 165 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 192, **USD 27 over the target**. Since v0.3 (USD 178) the total has risen by USD 14 for decisions of 2026-10-02: a spare load capacitor for selection (USD 6) and the keypad and status LEDs, BOM line 18 (USD 8: membrane keypad with printed overlay about USD 4.50, two LEDs with clips about USD 2, header and lead USD 1.50). The increase over v0.2 (USD 164) is the parts added to make the design buildable (PVT-DDR-003): the chassis plate (USD 3), the USB-C socket (USD 5), the pod ADC module (USD 3), the pod clip and gland (USD 1) and fixings (USD 2). Savings worth trying are listed in the design decisions register.

■ 8. Results

Table 6. Requirement status (from `docs/04-calcs/results.csv`)

ID	Value	Target	Status
R1	0 to 100 V; high-voltage case 83 V at -10 °C; capacitors 160 V, MOSFETs 150 V	0 to 100 V, below 120 V	Met (design review)
R2	0 to 20.48 A full scale; shunt 1.6 W of 3 W at 20 A	0 to 20 A	Met (design review)
R3	20.1 ms high-current case with the bank selected to 6.45 mF (19.9 ms at the decided 6.4 mF); 32 ms reference; up to 81 ms	20 to 200 ms	Met on paper with a selected bank (PVT-DEC-001 item 4)
R4	501 pairs minimum (high-current case, selected bank 6.45 mF)	200 or more	Met
R5	V $\pm 0.17\%$ $\pm 0.024\%$ FS; I $\pm 0.21\%$ $\pm 0.026\%$ FS	$\pm 1\%$ of reading $\pm 0.1\%$ FS	Met on paper (needs calibration)
R6	Instrument noise 0.08 % per sample	Pmax $\pm 1\%$ over 3 sweeps	Not verifiable at TRL 3
R7	Irradiance ± 3.6 to $\pm 5.4\%$; temperature $\pm 2\text{ }^{\circ}\text{C}$ budget	$\pm 5\%$, $\pm 2\text{ }^{\circ}\text{C}$	At risk
R8	± 3.7 to $\pm 5.4\%$ (needs reference cell within $\pm 4.5\%$)	$\pm 5\%$ or better	At risk
R9	About 7 or more samples per volt; rules drafted	Flags within 5 s	Not verifiable at TRL 3
R10	A, B, C, reject thresholds decided (PVT-DDR-002); editable table	Partner-editable grades	Met (design review)
R11	30 V in 0.21 s; 60 V passive in 40 s; opens below 0.5 A after up to 6 ms more	30 V in 2 s; 60 V in 60 s; open below 0.5 A	Met
R12	1.49 kg; 234 x 144 x 80 mm over bumpers	1.5 kg; 250 x 150 x 100 mm	Met
R13	Inside about 56 °C (light case) to 73 °C (dark) at 45 °C in sun; TFT with sun hood, readability unproven; IP54 case	0 to 45 °C, sun, IP54, readable	Not met (display); at risk (heat)
R14	10.3 h, about 1,853 sweeps	8 h or 200 sweeps	Met
R15	CSV fields defined in PVT-PRC-001; no firmware at TRL 3	CSV per sweep, USB or Wi-Fi export	Met (design review)
R16	No insulation test in this instrument	Detect insulation faults	Not met (out of scope)
R17	USD 192 in parts	Value-engineering target USD 165	Over the target by USD 27

Safety: These calculations cover a live DC instrument at up to 100 V and 20 A that stores up to 33 J. The discharge times assume both the active and passive paths are fitted and working; never rely on one alone, and treat the capacitors as charged until the display reads below 30 V. The heat results show the Li-ion cell can exceed its charging limit in sun; charge only in shade with the isolator open.